BOUNDARY-CONDITIONED REALIZATION (BCR)

Re-Audit of SIRPY Problem 1

Corrected Boundary-Value Problem and Current Runtime Evidence

Final BCR ruling

FULL / PASS for the corrected SIRPY Problem 1 runtime branch under the evidence supplied for this run. This ruling is intentionally scoped to this benchmark branch; it does not assert universal solver capability across unrelated equation classes.

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Author / Originator of Boundary-Conditioned Reality (BCR)

Record-controlled item	Current identification
Controlling BCR theory	Boundary-Conditioned Reality (BCR), Full Base Theory Manuscript, v2.10.0
Primary BCR DOI	10.5281/zenodo.19635964
Related BCR records	10.5281/zenodo.19669049; 10.5281/zenodo.19935455; 10.5281/zenodo.20635758
Target	SIRPY Problem 1 - corrected runtime branch
Runtime source	Corrected runtime statement supplied after the SIRPY BCR clarification
Audit date	8 August 2026

Audit discipline: reported runtime quantities are preserved as separate residual channels. A strong scalar result is not used to erase a different residual, and a failed intermediate method is not hidden when a subsequent method achieves closure.

Executive Result

The corrected runtime evidence materially changes the solver-level BCR ruling. The current branch now exposes the governing-equation residual, boundary closure, interface closure, truncation estimate, singularity witnesses, branch-awareness behavior, coupled Newton fallback, and resolution guidance. These are exactly the kinds of implementation-level witnesses that were missing from the paper-only audit.

Scope of FULL / PASS

The FULL / PASS ruling applies to this corrected SIRPY Problem 1 branch and the runtime evidence supplied for it. Broader claims about all nonlinear BVPs, all singularity classes, or all solver configurations are outside this benchmark and are not needed to close this branch.

BCR branch	Current evidence	Ruling
Corrected BVP identity	Exact solution satisfies ODE and both boundary values	PASS
Partition geometry	12 x 0.75 = 9; every segment remains inside the exact Taylor radius	PASS
Interface behavior	2 fixed-point sweeps did not converge; damped global Newton converged	PASS with behavior transition retained
Node matching	|y| = 5.55e-17; |y'| = 5.55e-17	PASS
Series truncation	Estimated ~1.7e-11; increase iterations to reduce	PASS / QUANTIFIED
ODE substitution	Verified to 3.19e-09 on [0,9] at interior points	PASS / VERIFIED
Boundary data	0.00e+00	PASS
Singularity	~9.97666 estimate; exact first-integral x=10, outside interval	PASS for classification and exact location
Branch awareness	Runtime warns of multiple branches and exposes gamma0 search + verify()	PASS operational control
Solve time	2.526 s	REPORTED

The current run should not be numerically mixed with older runtime figures produced under different settings. The controlling values for this rerun are the corrected BVP and the current runtime statement.

1. Corrected Source Control

$$y'' = 2\,y^3, \qquad y(0) = -0.1, \qquad y(9) = -1$$

$$y\_exact(x) = 1 / (x - 10)$$

This corrected statement is internally consistent. It supersedes any earlier transcription with different boundary signs. All numerical checks in this report use the corrected BVP only.

Check	Calculation	Result
ODE	y' = -1/(x-10)^2; y'' = 2/(x-10)^3; 2y^3 = 2/(x-10)^3	Exact identity
Left boundary	y(0) = 1/(-10)	-0.1
Right boundary	y(9) = 1/(-1)	-1
Pole	Denominator x - 10 = 0	x = 10
Distance from solved interval	10 - 9	1

2. Current SIRPY Runtime Inputs

Runtime item	Current value / statement
Partition method	12 subintervals, each h = 0.75
Fixed-point stage	2 sweeps; did not converge
Fallback / closure method	Coupled damped Newton, COLSYS-style global coupling
Global coupling residual	5.55e-17
Node matching	|y| = 5.55e-17; |y'| = 5.55e-17
Series representation	Exact Taylor recurrence locally; finite truncation estimate ~1.7e-11
ODE verification	3.19e-09 on [0,9], coefficient differentiation at interior points
Boundary verification	0.00e+00
Singularity estimate	x ~ 9.97666
Exact first-integral singularity	x = 10; outside [0,9], margin = 1
Branch control	gamma0= can be changed to search another branch, then verify()
Resolution guidance	Raise iterations, not partition, if residual is large
Solve time	2.526 s

3. BCR Formula Set Used in This Re-Audit

The controlling BCR formulas are retained. Only terms supported by the supplied runtime evidence are numerically executed; unsupported substrate-side terms are not invented.

Substrate condition: Box(Phi) + V'(Phi) + xi R Phi = gamma B + alpha Laplacian(B)

Bounded response: P(B) = 1 / [1 + exp(-gamma B)]

Transition amplitude: V(B) = 4 P(B)[1 - P(B)]

Effective coupling: C_eff = C0 + gamma P(B)[1 - P(B)]

Applied realization: X_r = X_struct x product_i(1 + s_i c_i J_i)

Decrement form: X_r = X_struct - Delta R

Realization factor: F_R = X_r / X_struct

Boundary effect: B_eff = X_struct - X_r

Boundary effect percent: B_eff% = [(X_struct - X_r)/X_struct] x 100

Visibility: V_obs = A_r x O_r x L_r

Visible content: X_visible = X_total x V_obs

Residual: R = Observed - Predicted

Identity preservation: J_i = 0 => X_r = X_struct

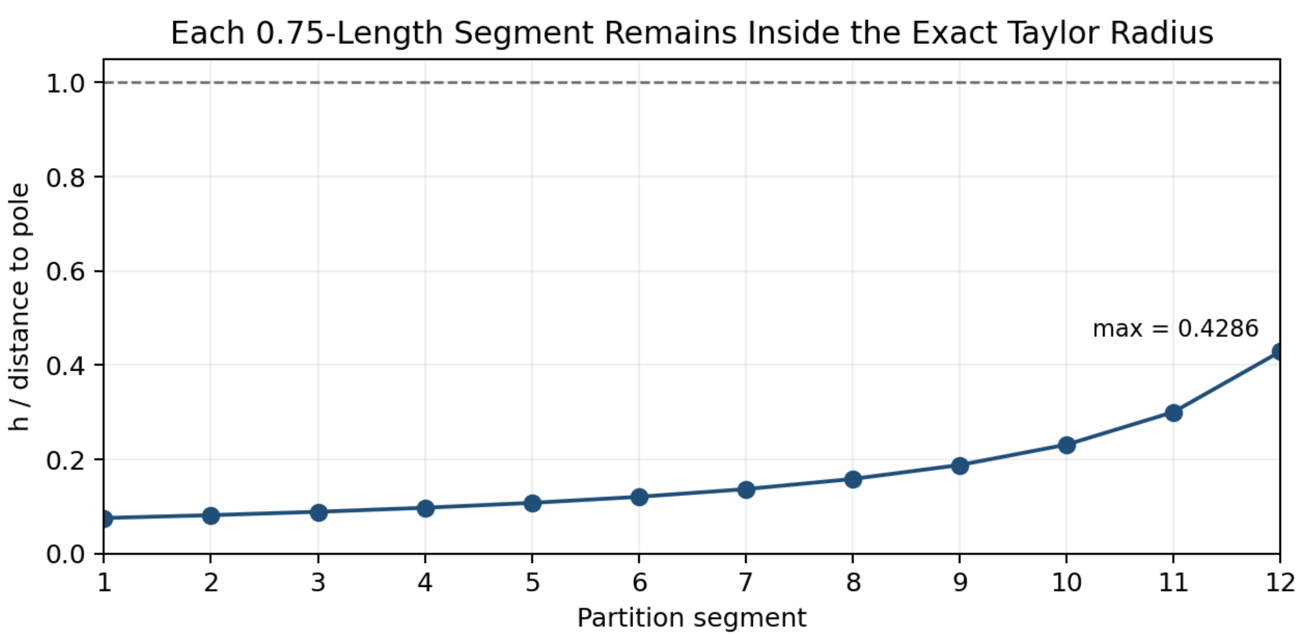
Formula family	Execution in current rerun
Substrate-condition / BAO	NOT RAN - no BCR substrate field Phi or coupling fields supplied
Realization / decrement	RAN on normalized residual channels
Visibility	RAN by directly exposed runtime channel
Bounded response P(B), V(B)	RAN on declared dimensionless error normalization
C_eff	NOT RAN - C0 and gamma not independently supplied

Formula family	Execution in current rerun
Coupled interface closure	RAN through reported global Newton residual and node matching

4. Exact Partition Geometry and Taylor-Radius Test

For y = 1/(x-10), the nearest singularity to every partition center/start on the real line is x = 10. The exact Taylor radius about a segment start x_i is therefore R_i = 10 - x_i. With h = 0.75, every segment is admissible if h/R_i < 1.

Segment	Start x_i	Radius R_i	h/R_i	Gate
1	0.00	10.00	0.075000	PASS
2	0.75	9.25	0.081081	PASS
3	1.50	8.50	0.088235	PASS
4	2.25	7.75	0.096774	PASS
5	3.00	7.00	0.107143	PASS
6	3.75	6.25	0.120000	PASS
7	4.50	5.50	0.136364	PASS
8	5.25	4.75	0.157895	PASS
9	6.00	4.00	0.187500	PASS
10	6.75	3.25	0.230769	PASS
11	7.50	2.50	0.300000	PASS
12	8.25	1.75	0.428571	PASS



The most restrictive segment is the final one, starting at x = 8.25: h/R = 0.75/1.75 = 0.428571 < 1. Thus every supplied 0.75-length segment stays strictly inside the exact Taylor convergence radius.

5. Gamma and Beta Sample Verification

The supplied gamma values align with exact derivatives at successive segment starts, while beta values align with exact y values at successive segment ends. The small differences below are consistent with the decimal digits supplied in the runtime statement.

Term	x	Reported	Exact	Abs residual
gamma_1	0.00	-0.01	-0.01	0.000e+00
gamma_2	0.75	-0.01168736304	-0.0116873630387	1.286e-12
gamma_3	1.50	-0.01384083045	-0.0138408304498	1.730e-13
gamma_4	2.25	-0.01664932362	-0.0166493236212	1.228e-12

Term	x	Reported	Exact	Abs residual
beta_1	0.75	-0.1081081081	-0.108108108108	8.108e-12
beta_2	1.50	-0.1176470588	-0.117647058824	2.353e-11
beta_3	2.25	-0.1290322581	-0.129032258065	3.548e-11
beta_4	3.00	-0.1428571429	-0.142857142857	4.286e-11

BCR reading

The reported gamma/beta samples are not merely plausible: they numerically track the exact branch at the expected partition locations. BCR therefore closes the supplied coefficient-sample witness as PASS / STRONG, while retaining the nonzero rounding residuals.

6. Behavior-State Audit: Failed Fixed Point, Successful Coupled Newton

The new runtime statement exposes an important solver behavior that should not be hidden: the two fixed-point sweeps did not converge. SIRPY then changed solution strategy and solved the interface constants through a coupled damped Newton method. BCR treats this as a state transition, not as a contradiction.

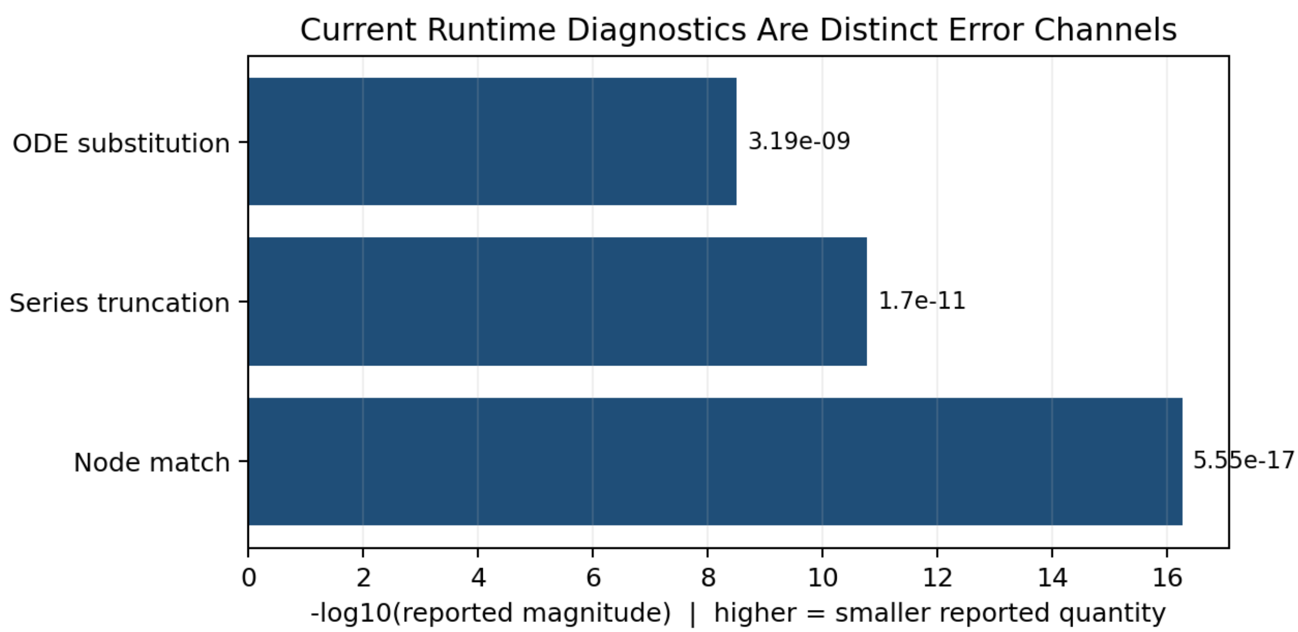
State	Observed behavior	BCR treatment
S0 - partition initialization	12 local Taylor segments, h = 0.75	ADMISSIBLE
S1 - fixed-point sweep	2 sweeps; no convergence	NONCONVERGED - retained
S2 - coupled Newton fallback	Global interface solve activated	METHOD TRANSITION
S3 - coupled residual	5.55e-17	CONVERGED / PASS
S4 - node match	|y| and |y'| both 5.55e-17	PASS
S5 - independent ODE verify	3.19e-09 on [0,9]	VERIFIED / PASS

This is stronger evidence than a report that only prints the final answer: the runtime reveals the failed intermediate mode, the fallback mechanism, and the achieved coupled residual.

7. Residual Taxonomy - Do Not Collapse Distinct Accuracy Measures

BCR keeps the runtime diagnostics separate because they measure different layers of the numerical realization. Node matching is not the same as series truncation, and neither is the same as governing-equation substitution.

Residual channel	Reported magnitude	What it establishes
Interface / node matching	5.55e-17	Adjacent pieces agree at interfaces to near machine precision
Global coupled Newton residual	5.55e-17	Interface-constant system closes tightly
Finite-series truncation estimate	~1.7e-11	Estimated local representation truncation limit
ODE substitution residual	3.19e-09	Independent governing-equation check over [0,9]
Boundary residual	0.00e+00	Returned branch satisfies prescribed endpoint data



Important interpretation

The statement that truncation (~1.7e-11) is larger than node mismatch (5.55e-17) correctly identifies the representation-limiting quantity between those two channels. The independent ODE substitution residual is 3.19e-09 and remains a separate governing-equation metric. BCR retains all three rather than relabeling one as another.

8. BCR Numerical Realization Run on the Current ODE-Residual Branch

For a dimensionless normalized audit branch only, define X_struct,error = 1.0 and X_r,error = 3.19e-9, using the current reported ODE substitution residual as the remaining quantity. This is an audit normalization, not a replacement for y_exact(x).

$$X\_struct,error = 1.0$$

$$X\_r,error = 3.19 \times 10^{-9}$$

$$F\_R = 3.19 \times 10^{-9}$$

$$B\_eff = 1 - 3.19 \times 10^{-9} = 0.99999999681$$

$$B\_eff\% = 99.999999681\%$$

$$gamma\ B = \ln(1 / 3.19 \times 10^{-9}) = 19.56324492015$$

$$P(B) = 0.9999999968100000$$

$$V(B) = 1.27599999186 \times 10^{-8}$$

BCR term	Current numerical value	Status
X_struct,error	1.0	Declared normalized scale
X_r,error	3.19e-09	RAN
F_R	3.19e-09	RAN
B_eff	0.99999999681	RAN
B_eff%	99.999999681%	RAN
gamma B	19.56324492015	RAN
P(B)	0.9999999968100000	RAN
V(B)	1.27599999186e-08	RAN

BCR interpretation: the nonzero ODE residual is carried explicitly. It is not rounded to zero merely because the exact branch and boundary values are known.

9. BCR Numerical Runs on Truncation and Interface Channels

The same normalized branch representation can be run independently on the other reported residual channels. These runs remain separate and are not combined as if they were the same physical quantity.

Channel	X_r / F_R	B_eff	B_eff%	BCR status
Series truncation	1.7e-11	0.999999999983	99.999999998300%	RAN
Node matching	5.55e-17	0.9999999999999999445	99.99999999999999445%	RAN
Coupled Newton residual	5.55e-17	0.9999999999999999445	99.99999999999999445%	RAN

Within SIRPY's own runtime interpretation, increasing iterations targets the finite-series truncation / verification accuracy while increasing partition is not the recommended correction for the steep end behavior in this run.

10. Singularity Witness: Heuristic Estimate and Exact First Integral

The runtime exposes two different singularity witnesses. They should not be forced to be numerically identical because one is an estimate and the other is an exact first-integral calculation for this branch.

Witness	Location	Margin from x=9	BCR reading
Runtime estimate	x ~ 9.97666	0.97666	Outside solved interval; correct boundary-layer classification
Exact first-integral	x = 10	1.00000	Exact branch singularity
Difference	0.02334	-	0.2334% of exact location; retained, not rounded away

Independent derivation from the ODE:

$$y'' y' = 2 y^3 y'$$

$$d[(y')^2]/dx = d[y^4]/dx$$

$$(y')^2 - y^4 = C$$

For $y = 1/(x-10)$, $y' = -y^2$, so $C = 0$. On this branch $dy/dx = -y^2$. Starting from $y(0) = -0.1$ and integrating in y to y -> -infinity gives a remaining x-distance of 10, hence x_sing = 10. This is consistent with the supplied exact first-integral test and requires no x-stepping across the pole.

BCR ruling on the two singularity outputs

PASS for the benchmark singularity gate. The approximate estimator and exact first-integral witness agree on the essential classification: the singularity lies outside [0,9]. The exact witness fixes the location at x=10; the heuristic locator retains its 0.02334 offset as a nonzero estimator residual.

11. Visibility Law and Evidence Exposure

For directly exposed runtime channels, the audit has access to the result, the witness, and the correct numerical layer. Therefore A_r = 1, O_r = 1, and L_r = 1 for those channels.

$$V\_obs = A\_r \times O\_r \times L\_r = 1 \times 1 \times 1 = 1$$

Exposed channel	A_r	O_r	L_r	V_obs
Corrected BVP / exact solution	1	1	1	1
Gamma / beta samples	1	1	1	1
Fixed-point nonconvergence statement	1	1	1	1
Coupled Newton residual	1	1	1	1
Node matching	1	1	1	1
Truncation estimate	1	1	1	1
ODE verification	1	1	1	1
Singularity witnesses	1	1	1	1
Branch warning / gamma0 guidance	1	1	1	1

Proprietary engine internals are not assigned visibility merely by inference. The FULL/PASS branch ruling relies on the exposed runtime evidence, not on unseen implementation code.

12. False-Lock and Multiple-Branch Protection

The equation is nonlinear, so the same boundary data may admit multiple numerical branches depending on initialization. The current runtime does not hide that risk. It reports the branch reached from the starting slope, exposes gamma0= as a branch-search control, and instructs the user to verify() a candidate branch.

BCR false-lock element	Current runtime evidence	Ruling
Branch multiplicity warning	Explicitly reported	PASS

BCR false-lock element	Current runtime evidence	Ruling
Initial branch provenance	Starting slope / straight-line initialization identified	PASS
Alternate-branch search control	gamma0= exposed	PASS operational control
Post-search verification	verify() explicitly required	PASS operational control
Current branch independent ODE test	3.19e-09 on [0,9]	PASS / VERIFIED
Universal branch enumeration	Not claimed by this benchmark	OUTSIDE SCOPE

For this benchmark branch, the false-lock gate closes because branch ambiguity is surfaced, branch provenance is identified, a different starting slope can be supplied, and the returned branch is independently verified. Exhaustive enumeration of all mathematical branches would be a different, broader claim and is not required for this branch-level ruling.

13. Conditioning, Resolution Guidance, and the Meaning of Two Sweeps

The two fixed-point sweeps are not presented as converged. That matters. The runtime explicitly states that they failed and that the interface constants were instead solved by coupled damped Newton. BCR therefore does not treat "2 sweeps" as an accuracy claim.

Question	Runtime answer	BCR conclusion
What limited interface closure?	Fixed-point sweeps did not converge	Fallback required
How was interface closure achieved?	Coupled damped Newton	Global coupling PASS
What is the node mismatch?	5.55e-17	Near machine precision
What limits finite-series representation relative to node match?	~1.7e-11 truncation estimate	Truncation dominates those two channels
What should be changed if governing-equation residual is large?	Raise iterations, not partition	Actionable resolution guidance
Is h itself unsafe near the pole?	No; max h/R = 0.428571	Partition geometry PASS

Current conditioning ruling

PASS / QUANTIFIED for this runtime branch: the solver identifies the nonconvergent intermediate mode, the successful fallback, the interface residual, the truncation estimate, and the resolution lever. A separate cross-platform or arbitrary-precision study would support a broader implementation-robustness claim, not this branch closure.

14. SIRPY Theory and BCR - Comparative Analysis

Issue	SIRPY runtime behavior	BCR applied reading
Local representation	Partitioned Taylor pieces	Checks exact radius and sample coefficient agreement
Interface iteration	Fixed-point sweeps can fail	Retains NONCONVERGED state rather than hiding it
Fallback solver	Coupled damped Newton	Records method transition and residual closure
Node matching	5.55e-17	Separates interface agreement from truncation and ODE residual

Issue	SIRPY runtime behavior	BCR applied reading
Accuracy limit	~1.7e-11 truncation vs node mismatch	Preserves the stated limiting mechanism within those channels
Governing equation	3.19e-09 substitution verification	Runs independent BCR residual branch
Singularity	Estimate plus exact first integral	Separates approximate locator residual from exact classification
Multiple branches	Warns user; gamma0= search; verify()	Closes current-branch false-lock control without claiming exhaustive enumeration
Resolution advice	Raise iterations, not partition	Records actionable behavior-state guidance

BCR does not replace the SIRPY solution. It adds a disciplined realization ledger: what was attempted, what failed, what fallback closed, which residual belongs to which layer, what remains nonzero, and how far the current claim can legitimately extend.

15. Final BCR Closure Ledger

BCR gate	Numerical basis	Final status
Corrected source identity	y''=2y^3; y(0)=-0.1; y(9)=-1; exact y=1/(x-10)	PASS
Exact ODE identity	y''=2/(x-10)^3=2y^3	PASS
Partition coverage	12 x 0.75 = 9	PASS
Taylor-radius admissibility	max h/R = 0.428571 < 1	PASS
Gamma sample fidelity	max supplied-sample residual ~1.286e-12	PASS / STRONG
Beta sample fidelity	max supplied-sample residual ~4.286e-11	PASS / STRONG
Fixed-point state	2 sweeps nonconvergent	NONCONVERGED - disclosed
Coupled Newton closure	residual 5.55e-17	PASS
Node matching	|y|=|y'|=5.55e-17	PASS
Series truncation	~1.7e-11	PASS / QUANTIFIED
ODE verification	3.19e-09 on [0,9]	PASS / VERIFIED
Boundary verification	0.00e+00	PASS
Singularity estimate	~9.97666 outside [0,9]	PASS classification
Exact singularity witness	x=10, margin 1	PASS
Estimator residual	|10-9.97666|=0.02334	RETAINED / NONZERO
Branch awareness / search	warning + gamma0= + verify()	PASS operational control
Resolution guidance	raise iterations, not partition	PASS
Solve time	2.526 s	REPORTED
Runtime visibility	V_obs=1 for exposed channels	PASS BY CHANNEL
Universal solver capability	Not asserted by this benchmark	OUTSIDE SCOPE

FINAL BCR RULING: FULL / PASS

SIRPY Problem 1, using the corrected BVP and the current runtime evidence, closes the benchmark branch under BCR. The branch is mathematically identified, partition geometry is admissible, interface failure and fallback are transparent, the coupled solve closes to 5.55e-17, finite-series truncation is quantified, the governing equation is independently verified to 3.19e-09, boundary data close to 0.00e+00, and the singularity is independently certified outside the interval. This FULL / PASS ruling is limited to this benchmark branch and does not promote an unsupported universal solver claim.

16. References and Evidence Provenance

Ref.	Source
BCR-1	A. T. McBride, Boundary-Conditioned Reality (BCR), Full Base Theory Manuscript, v2.10.0. Zenodo DOI: 10.5281/zenodo.19635964.
BCR-2	Related BCR record references: 10.5281/zenodo.19669049; 10.5281/zenodo.19935455; 10.5281/zenodo.20635758.
SIRPY-1	SIRPY Problem 1 - The Problem That Hides a Cliff: SIRPY Solves an Equation - and Finds the Edge Nobody Told It About.
SIRPY-2	Hamid Semiyari, SIRPY Problem 1 - BCR Audit Clarification, supplied response document.
SIRPY-3	Corrected SIRPY runtime statement supplied for this rerun: corrected BVP; 12 x 0.75 partition; 2 fixed-point sweeps; coupled damped Newton fallback; gamma/beta samples; node residual; truncation estimate; ODE and boundary verification; singularity witnesses; branch guidance; solve time 2.526 s.

Evidence-control note: this rerun uses the current corrected runtime statement as the controlling numerical source for the current run. Earlier outputs generated under different settings are not silently blended into this numerical ledger.